# ERNEST MIKHAIL SACDAL

Software Engineer | Full-Stack Developer | AI/ML Engineer
Bankstown, NSW 2200 | sacdalernest01@gmail.com | +61 468 933 638
ernestmikhail.com | linkedin.com/in/ernestmikhail | github.com/ernestsacdal

## PROFESSIONAL SUMMARY

Full-stack and AI engineer who builds and ships production LLM-powered and agentic systems end-to-end — FastAPI and Python on the backend, Next.js and TypeScript on the front. Two years of industry experience across multi-tenant platforms, RESTful APIs, and AI integrations, plus a portfolio of independently shipped products including an 8-agent code-migration pipeline and a price-prediction iOS app on the App Store. Currently completing a Master of Information Technology in Sydney, graduating Nov 2026 with full Australian work rights on graduation. Open to graduate, full-time, contract, and hybrid roles.

## TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript, SQL, PHP
**AI & LLM Engineering:** LangGraph, LangChain, Multi-Agent Systems, RAG, Async SSE Streaming, Claude API, OpenAI API, Groq, OpenRouter
**Backend:** FastAPI, NestJS, Node.js, Express.js, Django, REST API Design
**Frontend:** Next.js, React, React Native, Expo, Tailwind CSS
**Data & Infrastructure:** PostgreSQL, Supabase, MySQL, MongoDB, Docker, AWS, Vercel, Railway, Git, CI/CD
**Testing & Methods:** Pytest, Jest, Agile/Scrum

## EXPERIENCE

**Freelance Full-Stack Developer** *2025 – Present*
*Self-employed · Remote*
- Built a web application for an automotive repair business (Niks Prime Automotive) handling customer bookings, service tracking, and invoicing.

**Software Engineer** *Jan 2024 – Feb 2025*
*Lightweight Global Tech Solutions · Philippines*
- Built and owned full-stack development of multi-tenant utility-billing and ERP platforms (Express.js, Next.js, TypeScript).
- Developed a RESTful API in Django for a BPO locker system with automated device and attendance tracking, supporting 200+ daily active users.
- Built a beneficiary-management frontend in Next.js, improving interface responsiveness and usability.
- Prototyped an AI-powered proposal-generation agent integrating LLMs, reducing the team's manual drafting time.
- Optimised a customised insurance system, improving database query performance and reducing reporting inconsistencies.

**Freelance Full-Stack Developer** *2022 – 2024*
*Self-employed · Remote · USA, Philippines & Australia*
- Delivered custom web applications for clients using Django, Laravel, Express.js, and Next.js, owning each project end-to-end from requirements to handover.

## PROJECTS

**Mikhai — PHP-to-Laravel Migration Copilot** github.com/ernestsacdal/multi-agent-laravel-migrator
- 8-agent LangGraph pipeline that analyses legacy PHP codebases and generates Laravel migration plans with async SSE streaming. FastAPI, Qwen3-Coder and Llama 3.3 via OpenRouter and Groq.

**PreTriage — AI Patient Pre-Triage System** ernestmikhail.com
- Real-time triage assistant that turns patient-described symptoms into structured pre-triage output. Next.js and FastAPI with a Socket.IO real-time layer, Groq/Kimi K2 reasoning, and ElevenLabs voice.

**SQRLY — Multi-Tenant SaaS POS Platform**
- Multi-tenant point-of-sale SaaS built as a part-time developer at I-Dream IT Solutions, spanning a four-repository architecture (admin and merchant; frontend and backend). NestJS and Supabase with shared-schema multi-tenancy and row-level security.

**Wednesday — AI Grocery Price Prediction App** github.com/ernestsacdal/wednesday-ingestion
- iOS-first app that predicts Australian grocery special cycles using on-device AI via Apple Foundation Models. Expo, React Native, TypeScript, and Supabase.

## EDUCATION

**Master of Information Technology** *Mar 2025 – Nov 2026*
*Crown Institute of Higher Education · Sydney, NSW*
- Capstone: AI-based risk-scoring and rules engine, developed in collaboration with an industry partner.

**Bachelor of Science in Information Technology** *2020 – 2024*
*PHINMA Araullo University · Philippines*